

‘Speaking Brains’

Abstract of Monograph

Joseph Galasso

Joseph Galasso is a theoretical linguist at California State University, Northridge, who applies the "dual mechanism model" of human language to artificial intelligence (AI). He argues that a hybrid AI model is needed to replicate human thought, combining local-domain connectionist models (neural nets) with distant-domain symbolic models (rule-based reasoning). His recent research bridges theoretical linguistics and neuroscience to investigate how the brain processes language, with a particular focus on the roles of the basal ganglia and astrocytes. He compares these neural mechanisms to artificial intelligence (AI) models, arguing that biology provides a more complete model for understanding recursive language functions.

Basal ganglia and grammar

- **"Basal ganglia grammar":** Galasso posits that the basal ganglia, a subcortical region known for its role in motor control, also serves as a "common-denominator interface" for language.
- **Sequencing and recursion:** He suggests the basal ganglia's role in processing sequential actions extends to grammatical operations, particularly those that require deep, recursive embedding. This differs from the simpler "recurrent" patterns often found in AI models.
- **Dual Mechanism Model:** His work connects to the Dual Mechanism Model of language, proposing that the basal ganglia handle the rule-based, procedural aspects of language, such as complex grammatical structures.

Astrocytes and AI

- **Astrocyte-Ca²⁺ neuronal circuit:** Galasso emphasizes the computational significance of astrocytes, a type of glial cell, and their calcium (Ca²⁺) signaling. He proposes that the "Astrocyte-Ca²⁺ Neuronal Circuit" contributes to the slower, more enduring neural interactions necessary for complex language functions.
- **Beyond transformers:** He argues that current AI models, like transformers, fall short of replicating true human language abilities because they do not capture the unique dual-path synaptic interactions made possible by astrocytes. His research suggests that mimicking these intricate biological circuits is necessary to bridge the gap between AI and natural language.

Context within his broader work

Galasso's ideas on neural circuitry build on his earlier research on child language development, particularly the acquisition of syntax. In his book *Speaking Brains*, he synthesizes these topics, connecting child language acquisition, the basal ganglia, astrocytes, and AI. His work provides a neurobiological grounding for explaining the development of complex linguistic structures and the current limitations of AI language models.

Galasso's dual model for AI

Galasso's model draws on neuro-linguistic and psycho-linguistic research to propose that AI must incorporate two distinct computational processes, similar to the human brain, to achieve true human-like thought and language capabilities.

The two computational processes

- **Connectionist (or statistical) models:** These are neural network models that excel at pattern-matching and creating local, frequency-based associations, much like the human brain's hippocampus. They handle "item-based" learning, such as associating a word with another word based on how often they appear together. This is a fast and largely statistical process.
- **Symbolic (or rule-based) models:** These systems perform complex symbolic manipulation and recursive implementations, which are crucial for a deeper understanding of language and rules. Galasso equates this with the cortical processes in the human brain. This is a slower, deliberative process that handles abstract categories and structural dependencies.

The hybrid system

Galasso's core argument is that for an AI to truly understand and generate human language, it must have an interface that allows these two systems to work together,

mirroring the brain's cortico-hippocampal interplay.

- **Correlation vs. understanding:** Connectionist models can capture correlations based on statistics, but only symbolic models can provide the deeper, rule-based understanding that defines human language.
- **Balancing speed and depth:** This dual system would allow for the efficiency of fast, item-based associations and the robustness of slower, category-based reasoning.

Relation to current AI and Galasso's broader work

Galasso's research connects his linguistic and neurological framework to modern AI challenges, particularly with large language models (LLMs).

- **Beyond Transformer models:** His 2023 paper, *The Basal Ganglia, Astrocyte-Ca²⁺ Neuronal Circuit, and Artificial Intelligence*, specifically addresses the limitations of transformer models, arguing that an effective interface between AI and natural language requires a dual-mechanism approach.
- **Recursive syntax:** In his 2019 monograph *Recursive Syntax*, Galasso explored recursion as a core property of human language and its implications for AI, also incorporating discussions on the dual mechanism model.
- **AI and biolinguistics:** His work is part of a broader biolinguistics research program, cited by institutions like the NIH, which seeks to understand the biological basis of language and apply those findings to fields like AI and neuroscience.